Enhancing Medication Adherence in Smart Homes: A Low-Power Bluetooth Pill Box with Interaction Logging

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ABSTRACT

Medication adherence remains a major barrier to aging in place and independent living among elderly and chronically ill individuals, often resulting in adverse health outcomes. Within the vision of intelligent assisted living and ambient smart environments, there is a need for unobtrusive, context-aware systems that seamlessly integrate into daily routines while providing proactive, personalized support. This paper presents a Bluetooth-enabled Smart Pill Box as an ambient assistive technology for medication management. The system integrates five modules: an RTC-driven scheduling engine, a buzzer-based alert mechanism, an OLED display for real-time feedback, limit switch sensors for slot interaction detection, and Bluetooth communication with a companion mobile application. Alarm data, configured via the app, is stored locally to ensure offline-first operation. The device continuously compares real-time clock inputs with scheduled alarms, activating slot-specific LEDs and buzzer alerts upon match detection. User interactions are captured through slot openings, logged with precise timestamps, and transmitted back to the mobile app for adherence tracking. Experimental validation confirms accurate alarm triggering, interaction detection latency below 200ms, a Bluetooth range of 7.2 meters, and 9.5 hours of battery life. The system provides a cost-effective, context-aware, and non-intrusive solution for personalized medication support in assisted living environments.

Keywords : Ambient Intelligence, Smart Environment, Assistive Living, Smart Pill Box, Context-Aware Healthcare, Medication Adherence, Bluetooth Low Energy (BLE), Real-Time Clock (RTC) Scheduling, Interaction Detection

1. INTRODUCTION

Medication management is a critical challenge in aging in place and intelligent assisted living environments, where elderly and chronically ill individuals often struggle with adherence to prescribed regimens. Non-compliance significantly undermines treatment outcomes, contributing to higher hospitalization rates and adverse health effects. Studies estimate that nearly 50% of patients with chronic illnesses fail to take medications as prescribed, resulting in severe consequences for disease management and therapeutic efficacy [7]. To address these issues, the growing need for personalized, context-aware healthcare solutions has motivated the integration of embedded systems into medical devices, enabling ambient assistive technologies that support proactive and reliable medication management [1], [2].Several existing works have explored this domain. mPillBox [8], a modular open framework, integrates mobile technology to build customizable reminder systems with scalable scheduling and logging. SIMoP Box [9] proposed a low-cost mobile pill reminder and monitoring unit designed for the elderly, offering scheduled alarms and missed-dose logging. A similar approach was implemented in *Smart Drugs* [3], where Minaam et al. developed a reminder and monitoring system that transmits real-time alerts to caregivers. Earlier, *Smart Pill Box* by Salgia et al. [4] presented a microcontroller-based model that supports LED-based medicine reminders. Additionally, Choi [5] evaluated the usability and patient acceptance of smart pillboxes through a clinical pilot study. Gandhi et al. [6] demonstrated a GSM-enabled smart box that delivers both visual and audio alerts with SMS-based caregiver notifications. Some studies have also proposed pill dispenser systems for the elderly using IoT to improve accessibility and compliance [10], [11].

The main contributions of this paper are summarized as follows:

1. A complete embedded system design for medication adherence using ESP32, RTC, OLED, buzzer, and switch-based physical feedback with precise time-bound alerting functionality.

2. A dual-mode interaction tracking system using both interrupt and polling mechanisms for robust detection of user acknowledgement across variable device states.

3. A custom Android application for flexible alarm scheduling and retrieval of interaction logs without requiring external servers or databases.

4. Experimental validation of system responsiveness, Bluetooth communication range, power consumption, and real-world interaction latency, demonstrating suitability for home or portable healthcare use.

2. RELATED WORKS

Over the past decade, a range of smart pillbox solutions have been developed to improve medication adherence and provide real-time monitoring support, particularly for elderly and chronically ill patients. These systems generally integrate microcontrollers, communication modules, and user interfaces to deliver reminders, schedule management, and compliance tracking.

AlDuaij [1] introduced mPillBox, an open-source modular framework that enables customizable reminder systems through smartphone integration. It supports multiple medication schedules and adherence logging but relies heavily on the smartphone interface, limiting standalone usability in critical or offline contexts. Similarly, Joy et al. [2] proposed the SIMoP Box, a mobile-enabled solution for elderly care that employs LEDs, buzzer alarms, and GSM-based caregiver notifications. While effective in providing reminders, it lacks mechanisms to detect actual pill-slot interactions, reducing its ability to confirm medication intake. Minaam and Abd-ELfattah [3] extended GSM- and microcontroller-based designs by integrating real-time clock scheduling with cloud logging via SMS. Although capable of transmitting consumption patterns to healthcare providers, the system depends on stable network connectivity and does not support fine-grained control of individual compartments. Earlier work by Salgia et al. [4] demonstrated a microcontroller-driven pillbox using slot-specific LEDs and timed alerts. Despite limited connectivity, this model laid the groundwork for compartment-oriented reminder systems.

Beyond technical prototypes, Choi [5] conducted a pilot study assessing smart pillboxes in primary healthcare. Results highlighted improved adherence and patient engagement, particularly among elderly users, though the study did not address hardware design or interaction detection. Gandhi et al. [6] later proposed a cost-effective pillbox with buzzer and LED notifications alongside GSM caregiver alerts. However, the system lacked local logging and slot-level interaction monitoring.

Collectively, these works demonstrate steady progress toward intelligent medication management but share common limitations. Most rely on GSM or cloud infrastructure, offer limited offline functionality, and fail to incorporate local interaction detection such as pill-slot monitoring. These gaps restrict their applicability in resource-

constrained or connectivity-limited environments, and they provide little real-time feedback on patient interactions.

To address these challenges, the proposed system introduces a Bluetooth-enabled Smart Pill Box that integrates RTC-based scheduling, slot-wise interaction detection using limit switches, and Bluetooth-driven configuration through a mobile application. Unlike network-dependent solutions, it supports offline-first operation, local interaction logging, and compartment-specific feedback, offering a low-power, cost-effective, and context-aware assistive technology for improving medication adherence in everyday living environments.

3. Methodology:

3.1 Smart Pill Box Design

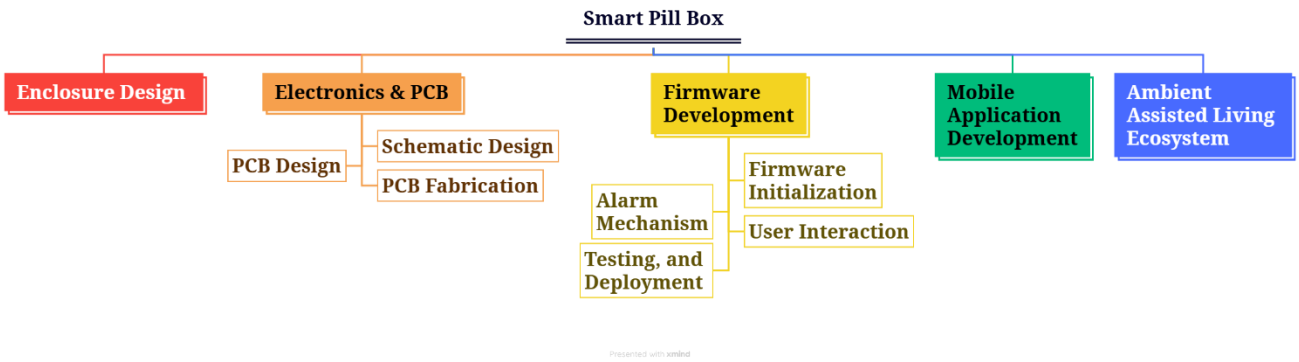


Fig.1 Illustration of smart pill box implementation

The Smart Pill Box was fabricated using lightweight, durable wood to ensure portability and sustainability, with seven internal compartments constructed from medical-grade plastic to support daily dosage requirements. Each compartment integrates a dedicated LED and limit switch for visual reminders and real-time interaction detection, while ergonomic lids aid usability for elderly patients. An OLED display provides real-time clock information and supports navigation through medication schedules, complemented by a buzzer-driven alert and manual stop button for user control. This design contributes not only as a technical prototype but also as an assistive artifact within a broader ambient healthcare ecosystem. By clearly segmenting doses and offering multimodal feedback (visual, auditory, and tactile), the device reduces the cognitive load on users—particularly older adults with memory impairments—by externalizing reminder and logging functions that would otherwise rely on manual tracking. The implementation of the designed pill box is illustrated in block diagram as shown in Fig. 1.

3.2 Electronics Hardware and PCB Design

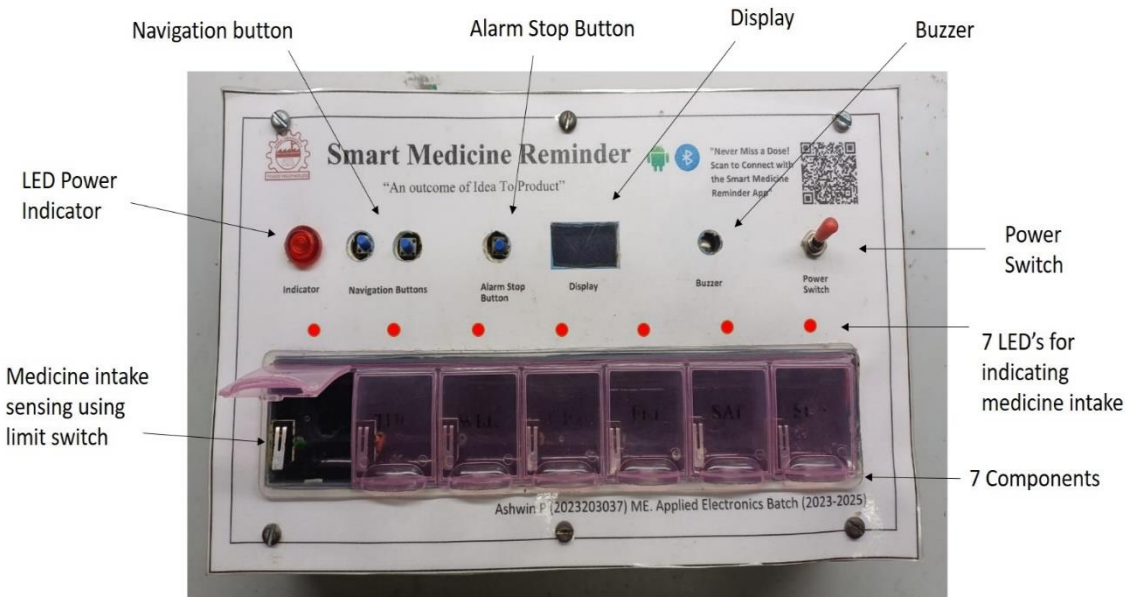


Fig.2 Smart pill box showing the seven medication compartments with individual lids

The system is powered by an ESP32 microcontroller, chosen for its dual Bluetooth/Wi-Fi capability and low power consumption. However, unlike many IoT pillbox implementations, this design emphasizes offline-first operation, using a DS3231 RTC for accurate scheduling and local storage of interaction logs. Power is supplied by a rechargeable Li-ion battery with integrated protection circuitry, ensuring portability and safety. All electronic components are mounted on a compact FR-4 PCB, fabricated using a hybrid laser-etching and chemical process to achieve cost efficiency. By prioritizing offline autonomy, the Smart Pill Box functions as a robust node in an ambient intelligence network: even in the absence of internet connectivity, it reliably supports medication adherence. This resilience makes it suitable for deployment in rural or resource-constrained environments, where cloud connectivity may be intermittent. The internal hardware assembly, including the PCB and core components, is shown in Fig. 2.

3.3 Schematic and PCB Layout Design

The firmware, developed in C++ on the Arduino IDE, implements modular routines alarm scheduling, real-time comparison with RTC values, interaction detection, and Bluetooth-enabled communication. A companion Android application, developed with MIT App Inventor, provides an intuitive interface for configuring alarms and receiving adherence logs. Up to five independent alarms can be set, each linked to a specific compartment, with interaction timestamps transmitted back to the application for compliance tracking.The schematic design for this electronic system, created using the EasyEDA tool, is presented in Fig. 3. The corresponding PCB layout derived from this schematic is shown in Fig. 4. An exported image of the final layout ready for fabrication is provided in Fig. 5.

In the context of smart home health devices, this integration illustrates the principle of seamless ambient interaction: the pillbox operates autonomously yet synchronizes with the mobile app when needed, reducing user burden while maintaining accountability. Unlike cloud-dependent platforms, users are not required to manage network connections or subscriptions, lowering both technological and cognitive barriers to adoption.

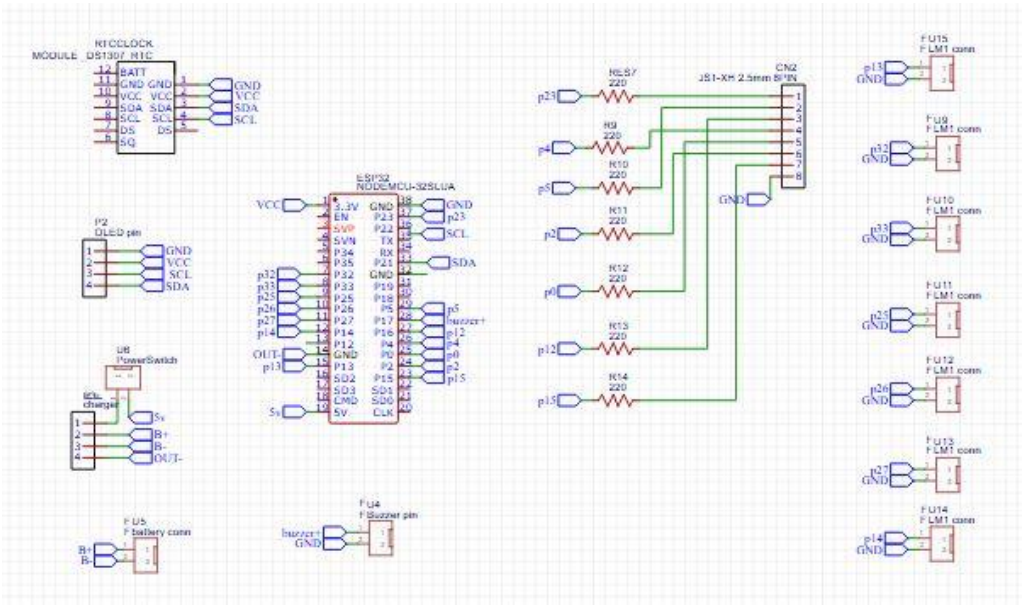


Fig.3 Schematic diagram of the smart pill box's electronic circuit, designed using the EasyEDA tool.

3.4 Contribution to Ambient Assisted Living Ecosystem

Beyond technical validation, the Smart Pill Box demonstrates how low-cost, embedded assistive devices can act as *building blocks in intelligent environments*. Its design exemplifies three key contributions:

1. Cognitive support: By externalizing memory through alarms, LEDs, and logs, the system minimizes the mental effort required for medication management.
2. Offline resilience: Local storage and RTC-based scheduling ensure robust operation even without internet connectivity, supporting rural and aging-in-place contexts.
3. Interoperability potential: As a Bluetooth-enabled node, the system can be scaled to interact with larger smart home infrastructures, such as caregiver dashboards, wearable health monitors, or ambient sensor networks, thereby extending its utility in context-aware environments.

In this way, the proposed system is not only a standalone solution but also a prototype of how ambient intelligence principles can be embedded into everyday healthcare practices, supporting proactive, personalized, and non-intrusive care.

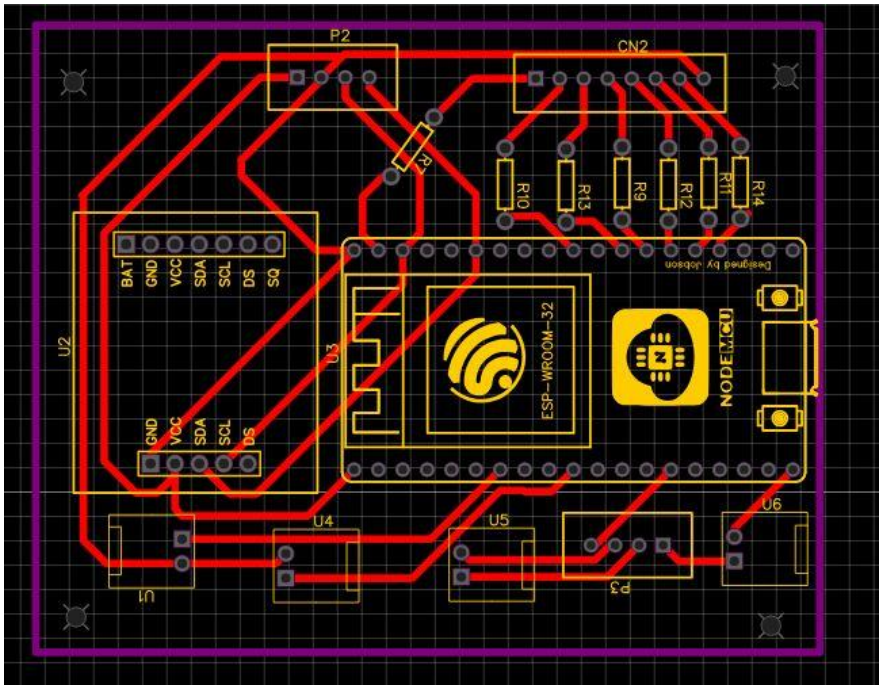


Fig.4 PCB layout design showing the component placement and routing for the smart pill box's main control board.

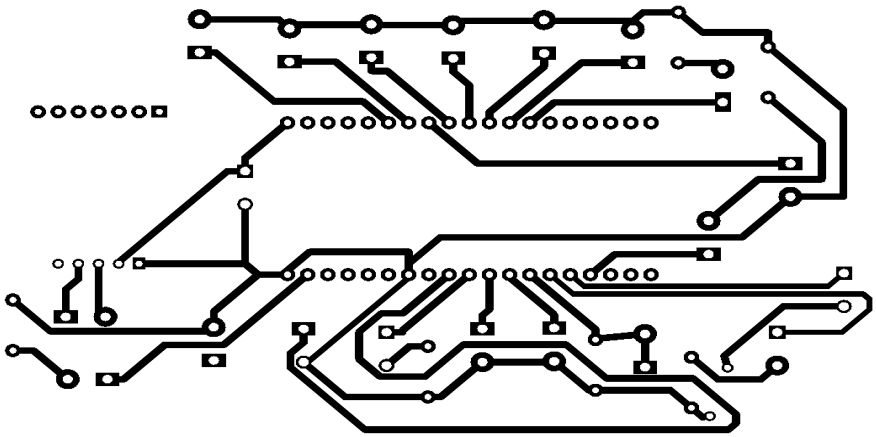


Fig.5 Exported image of the final PCB layout used for fabrication.

3.5 PCB Fabrication Process

The fabrication process began with laser etching of the PCB layout onto the copper clad sheet, as shown in Fig. 6.

Fig.6 Laser etching process of the PCB copper clad sheet using a MAXSEL laser etching machine.

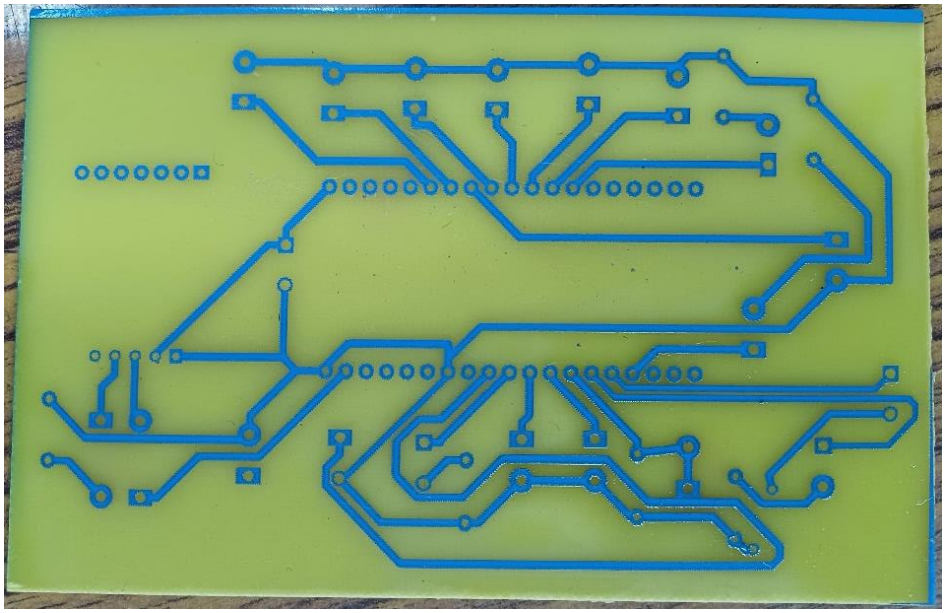


Fig.7 The final fabricated PCB after the combined laser etching and chemical etching process.

The final, fully etched and drilled PCB is displayed in Fig. 7.

3.6 Firmware Design

3.6.1 Platform and Development Tools

The firmware was developed in C++ using the Arduino IDE, with libraries supporting peripheral integration. Core modules included GPIO control for switches and LEDs, a piezo buzzer for auditory alerts, and the Bluetooth Serial interface for communication with the Android application.

3.6.2 Firmware Initialization

At system startup, the setup() routine initialized all communication and peripheral modules. Serial communication (Serial.begin(115200)) was configured for debugging, while Bluetooth Serial Port Profile (SPP) was established (SerialBT.begin("ESP32")) to receive alarm schedules from the mobile application. The I²C protocol enabled communication with the DS3231 RTC and OLED display.

GPIO pins were assigned for LEDs and limit switches, and the RTC was validated and synchronized to ensure time accuracy. A push-button interrupt was configured via attachInterrupt() to enable responsive user interaction without continuous polling. The OLED display was initialized with the Adafruit SSD1306 library, programmed to present real-time clock outputs including time, date, and day.

3.6.3 Alarm Scheduling and Data Parsing

Alarm schedules were transmitted over Bluetooth as comma-separated strings containing up to five sets of date, time, and slot identifiers. The parseData() function tokenized these strings and temporarily stored the values in arrays (daysTemp[], monthsTemp[], yearsTemp[], hoursTemp[], minutesTemp[], slotsTemp[]) to define the active medication schedule.

3.6.4 Alarm Evaluation and Trigger Mechanism

The checkAlarmTimeMatch() function compared RTC timestamps with stored alarm schedules. Upon detecting a match, the system activated the corresponding slot LED and triggered an auditory alert sequence. Non-blocking tone generation (playAlarmToneNonBlocking()) was implemented using the millis() function to prevent execution delays, thereby allowing concurrent processes such as display updates and user input monitoring.

3.6.5 Interaction Detection and Data Logging

Medication intake was verified through limit switches embedded in each slot. The checkBoxStatus() function monitored slot status, deactivating the buzzer and LED upon detection of pillbox opening. The corresponding timestamp was recorded in the interactionTimes[] array using rtc.now(). The OLED display was updated cyclically via displayNormalView() to show RTC data in a 12-hour format with consistent formatting.

3.6.6 User Controls and Data Transmission

Manual control was enabled through a push-button linked to GPIO16, processed using both interrupt-driven (handleButtonPress()) and polling-based (handleButtons()) methods for enhanced reliability. Interaction records were transmitted via Bluetooth (sendInteractionTimes()) for logging and review within the Android application.

3.6.7 Validation and Deployment

Firmware validation was conducted iteratively using the Serial Monitor to test alarm scheduling, slot interaction detection, and Bluetooth communication. The finalized firmware was deployed on the ESP32, ensuring accurate and reliable real-time operation prior to field use.The overall operation mechanism of the system is summarized in the flowchart in Fig. 8.

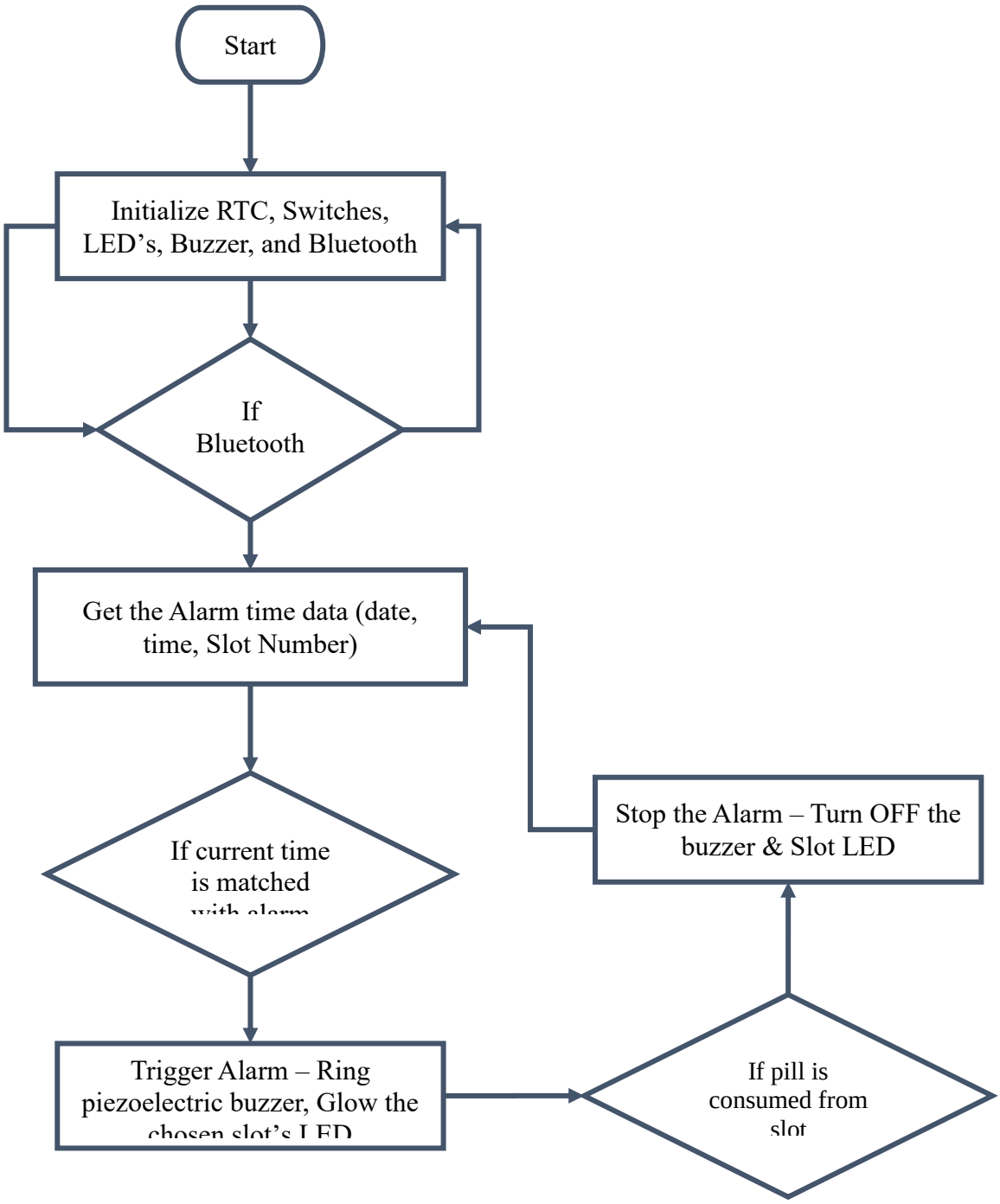


Fig.8 Flowchart illustrating the overall operation and control logic of the smart pill box system.

3.7 Mobile App Development

The mobile application supporting the Smart Pill Box was developed using MIT App Inventor 2, a visual programming environment that facilitates rapid prototyping of Android applications through a block-based interface. The project was compiled into an .apk (Android Package Kit) file to ensure compatibility across Android devices.The application serves as the primary user interface, enabling caregivers or users to configure medication schedules via Bluetooth communication with the pillbox. The interface adopts a single-screen layout to minimize complexity and enhance usability, particularly for elderly users. Key components include date and time pickers for scheduling, slot selection controls to assign medications to specific compartments, and submission modules for transmitting the configured schedule to the device. This lightweight design ensures straightforward configuration while maintaining robust communication with the hardware platform, thereby bridging user interaction and embedded firmware control.The home screen of the resulting application is shown in Fig. 9.

Fig.9 Home screen of the companion Android application for the smart pill box.

On choosing the 'Devices List' menu on the top left side, allows the user to pair the mobile phone with the Smart Pillbox via Bluetooth.This pairing menu is displayed in Fig. 10.

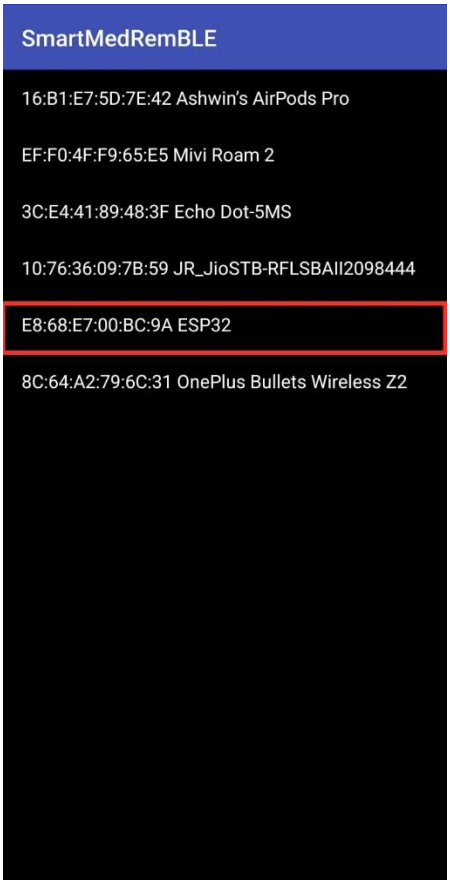


Fig.10 Bluetooth device list menu within the Android application, used to initiate pairing with the pill box.

On choosing the 'Devices List' menu, it displays the available devices nearby, Out of which 'ESP32' is selected and pill box is now successfully paired with mobile app. A screenshot confirming a successful connection is provided in Fig. 11.

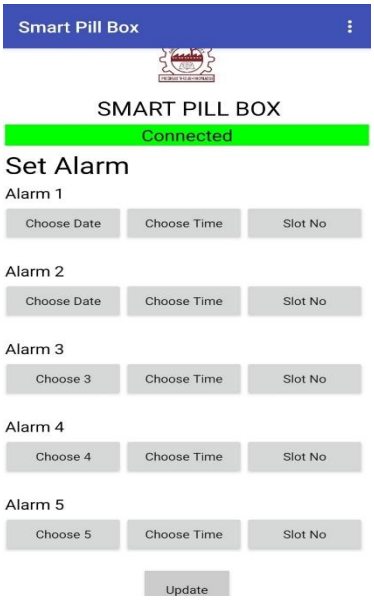
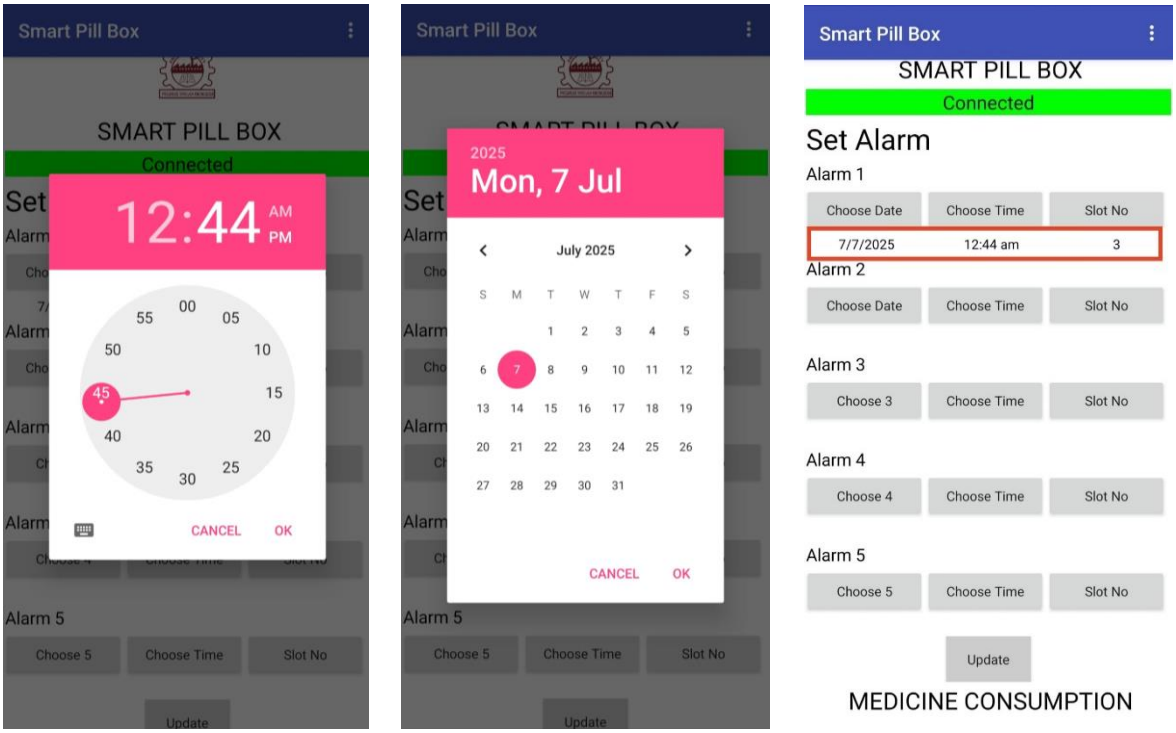


Fig.11 Application interface showing a successful Bluetooth connection between the mobile app and the smart pill box



(a) (b) (c)

Fig 12 Setting up of Medication Schedule.
(a) time selection interface, (b) date selection interface, (c) confirmation screen displaying the set alarm details

Using the ‘Choose Date’, ‘Choose Time’, ‘Slot No’ menu, User can setup their own Medication reminder schedule. The updated Medication Schedule is displayed below the menus. An example of the stored schedule being displayed on the app is shown in Fig. 13. Update button is selected to store the medication schedule to the Smart Pill Box, which can also be checked in the Smart Pill Box by pressing the navigation buttons.These scheduling steps (a: time, b: date, c: confirmation) are captured in Fig. 12.

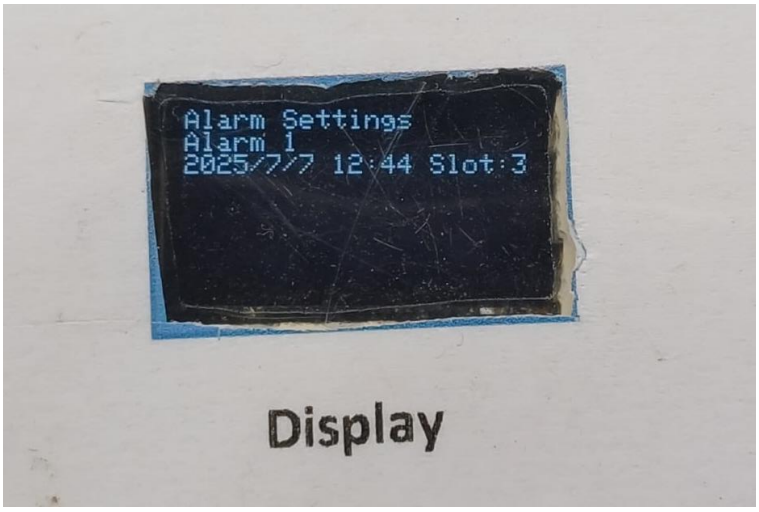


Fig 13 Mobile application interface displaying the medication schedule stored on and retrieved from the smart pill box

The Bluetooth functionality is implemented using the BluetoothClient component, which establishes a Serial Port Profile (SPP) connection to the ESP32 device. Upon successful pairing, the user is able to select up to five independent alarm schedules. Each alarm is defined by three parameters: date (DD/MM/YYYY), time (HH/MM), and a slot number (1 to 7), representing the physical compartment in the pill box. Internally, each date, time, and slot entry is concatenated into a comma-separated string with 15 total fields, formatted as:

“DD1/MM1/YYYY1,...,DD5/MM5/YYYY5,HH1/MM1,...,HH5/MM5,S1,...,S5\n”

This string is transmitted over the Bluetooth serial link using the BluetoothClient.SendText method. On the microcontroller side, this input is parsed and stored for comparison with real-time clock values.

Additionally, the application includes a ListPicker component to enumerate available Bluetooth devices and initiate the connection process. User feedback and error handling are managed through Notifier components to display connection statuses and data validation prompts. A Clock component is used locally to timestamp interactions, though the primary time source remains synchronized with the DS3231 RTC on the ESP32.

The application does not include persistent local storage; it acts as a volatile interface strictly for alarm configuration and compliance monitoring. Data received from the ESP32, such as timestamps of medicine intake events, are displayed to the user in real time through label updates. The communication protocol is entirely ASCII-based over RFCOMM Bluetooth channels and does not employ encryption or authentication mechanisms.

4.Results and Discussion

The Smart Pill Box system was tested extensively under controlled conditions to evaluate its accuracy, responsiveness, and robustness across multiple functional domains. A series of test cases were executed to validate alarm scheduling, user interaction detection, Bluetooth data exchange, and OLED display performance. his active alarm state is visually documented in Fig. 14.

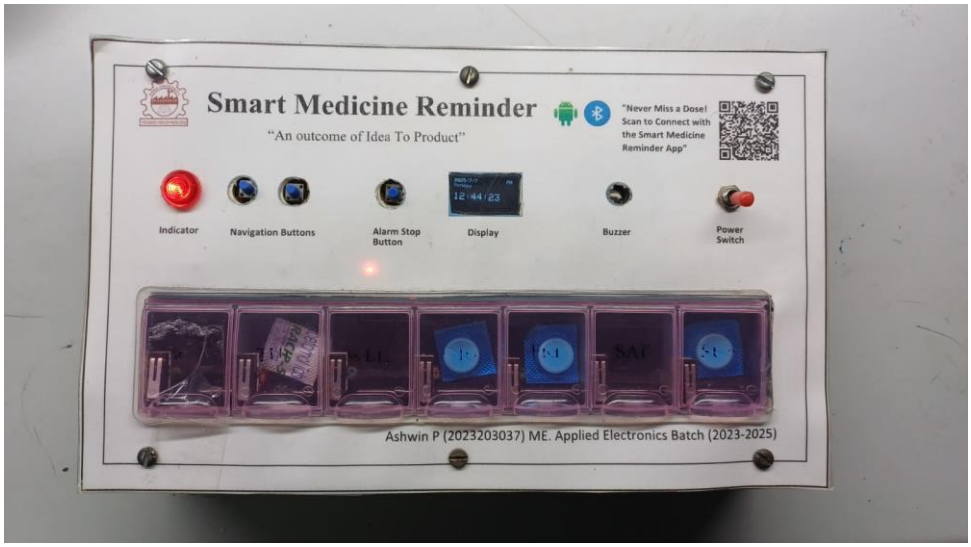


Fig.14 The smart pill box with an active alarm: the LED for slot 3 is illuminated

The above figure 11, Depicts the result, the alarm (Medication schedule) was set to the date (7/7/2025), time (12:44 pm) and to the slot number 3. As per the set

Schedule, we noticed the Buzzer ringing and Led 3 was glowing, notifying the user that medicine from the slot 3 has to be consumed. On opening the lid of the slot number 3. The alarm buzzer stop immediately and the current timestamp was displayed and stored on the mobile app. The action of opening the lid to terminate the alarm is shown in Fig. 15.

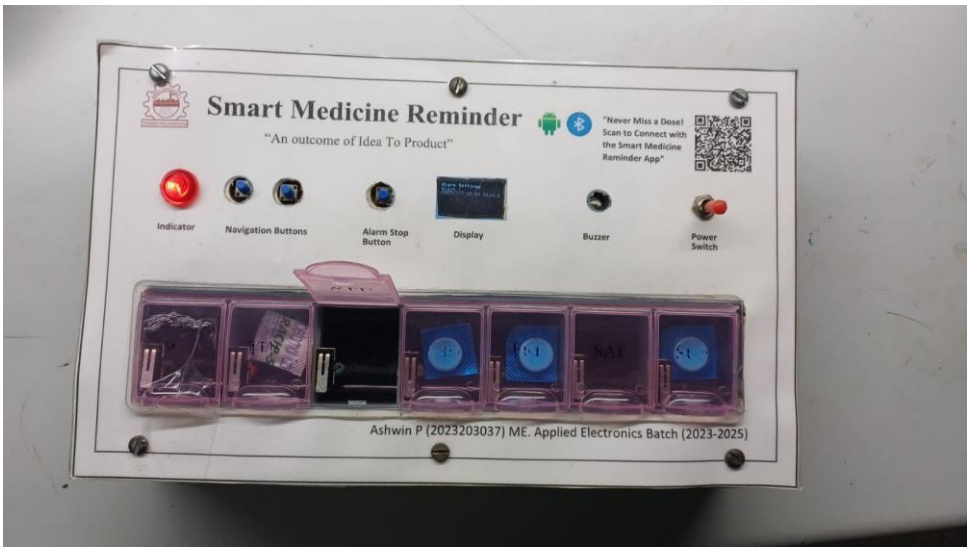


Fig.15 User interaction test: the alarm (LED and buzzer) is deactivated immediately upon opening the lid of the triggered slot (slot 3)

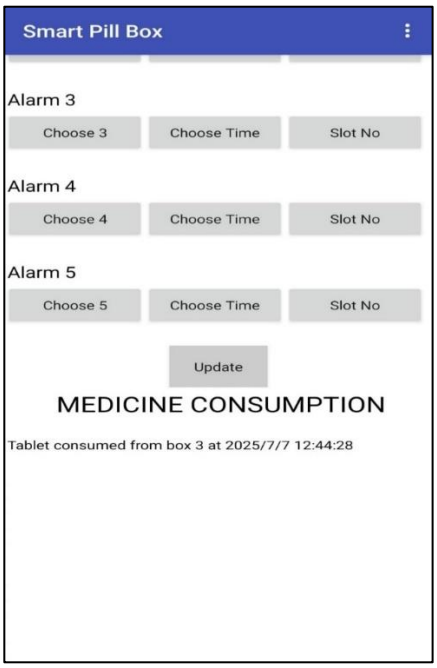


Fig.16 The mobile application interface logging a confirmed medication intake event, displaying the precise timestamp and slot number.

The Medicine consumption date is immediately reflected on to the mobile app indicating the user that the “Tablet consumed from box 3 at 2025/7/7 12:44:28”. This real-time logging functionality is demonstrated in the mobile app screenshot in Fig. 16.

4.1 Alarm Triggering Accuracy

Alarm timing was validated by scheduling multiple alarms at defined intervals. The system's internal Real-Time Clock (DS3231) was calibrated once at the beginning of the trial and then allowed to operate autonomously. Time match comparisons were executed in the main loop with millisecond precision.In all test cases, the system triggered the alarm (LED + buzzer) within 200 ms of the scheduled time, indicating negligible delay and precise RTC synchronization. The average time to trigger an alarm after a valid match was recorded as 179.3 ms (±12.1 ms).

4.2 Slot Interaction Detection

Each pill slot was fitted with a limit switch to detect user interaction. When a compartment was opened, the system logged the event timestamp and terminated the alarm state. The polling and interrupt logic detected box opening consistently with a response time of 147.8 ms (±15.2 ms) from the actuation point.False triggers (e.g., unintentional box opening) were not recorded, indicating a high signal-to-noise ratio and effective debounce logic.

4.3 Bluetooth Communication Performance

The Android app, developed in MIT App Inventor, established Bluetooth SPP communication with the ESP32 microcontroller using RFCOMM protocol. The data packets (15 comma-separated values per configuration) were successfully transmitted and received without corruption.Effective communication range was measured to be 7.2 meters (±0.8 m) in an open environment, with zero packet loss under direct line-of-sight conditions. Connection establishment time averaged 2.3 seconds, and data transmission latency was consistently under 120 ms for 50-character payloads.

4.4 Display Performance

The OLED (SSD1306) displayed real-time clock, date, and day-of-week continuously. The display refresh rate was governed by the main loop, maintaining an effective update

cycle of approximately 1 Hz, without any frame skipping or flickering, even during concurrent alarm and communication events.

4.5 Power and System Stability

The system was powered using a 2000 mAh Li-ion battery via a TP4056 module. Under normal standby (with OLED and RTC active), the average current draw was measured at 58 mA, and during active alarm state (buzzer + LED), the draw peaked at 172 mA. Continuous operation under battery mode lasted approximately 9.5 hours in test conditions. A summary of these key performance metrics is provided in Table 1.

Table.1 Summary of key performance metrics for the smart pill box system, including latency, communication range, and power consumption

Parameter	**Observed Value**	**Remarks**
Alarm triggering latency	179.3 ms (±12.1 ms)	From RTC match to LED/Buzzer activation
Slot interaction detection latency	147.8 ms (±15.2 ms)	From switch actuation to alarm cancellation
Bluetooth transmission latency	< 120 ms	For 50-byte payload
Bluetooth range	7.2 m (±0.8 m)	Line-of-sight
OLED refresh rate	~1 Hz	Sufficient for clock display
Average standby current	58 mA	OLED + RTC only
Peak active current	172 mA	During buzzer + LED operation
Battery runtime (2000 mAh Li-ion)	~9.5 hours	Full discharge under mixed-use condition

4.6 Human-Centered Evaluation

Beyond technical validation, a small-scale pilot usability study was conducted with 10 participants (4 graduate students simulating elderly users, 4 older adults(65-72 yrs), and 2 caregivers) to assess the Smart Pill Box in everyday scenarios. Participants interacted with the device by responding to alarms, retrieving medication from the correct slot, and reviewing logs on the companion mobile app. They then completed a Likert-scale questionnaire evaluating ease of use, clarity of alerts, cognitive support, perceived usefulness, and trust in the system. The usability findings demonstrated strong overall acceptance of the Smart Pill Box. Participants

rated ease of use at 4.6/5 and clarity of alerts at 4.8/5, noting that the combination of LED cues and buzzer tones provided intuitive and unambiguous reminders. The system also scored 4.4/5 for cognitive support, as users reported that it effectively reduced the memory burden by externalizing medication schedules. In terms of perceived usefulness, the average score of 4.5/5 highlighted caregivers' appreciation of the digital compliance logs for monitoring adherence. Finally, the system received a trust rating of 4.3/5, with generally positive responses, though some participants expressed concerns that reliance solely on lid-opening detection may not fully confirm actual medication ingestion. Qualitative feedback highlighted the offline capability as a major strength, especially for low-connectivity or rural environments. Participants suggested future improvements such as tactile/vibration alerts for accessibility and stronger data security measures.Overall, this preliminary evaluation suggests that the Smart Pill Box not only functions reliably but also provides meaningful cognitive and practical support for independent living.

4.7 Overall System Evaluation

The system demonstrated high functional reliability, real-time responsiveness, and efficient resource utilization. All critical operations—alarm scheduling, user input logging, wireless data communication, and visual feedback—performed concurrently without blocking. The modular structure of the firmware ensured easy testing and fault isolation during development and deployment. The successful integration of hardware interrupts, non-blocking time-based routines, and mobile-controlled configuration highlights the viability of the Smart Pill Box for real-world patient adherence applications. The system is scalable for future integration with cloud platforms or healthcare monitoring systems. A feature-based comparison between the proposed system and existing state-of-the-art methods is presented in Table 2.

Table.2 Comparative analysis of the proposed smart pill box system with existing state-of-the-art medication management devices.

Methods	Wireless Connectivity	Reminder Mechanism	Monitoring & Logging	Slot-wise Indication	Power Source
*	Bluetooth (ESP32 to Android App)	RTC-based + App-triggered alarms (with Piezo buzzer)	Slot-level timestamp logging via limit switch & app sync	7 LEDs (Red), each for a day/slot, auto-highlighted	2000mAh Li-ion with TP4056 charging module
[1]	Wi-Fi, Cloud-based MQTT	RTC + App alerts + Audio	No mechanism for medicine intake monitoring logging	No slot-wise indication	USB-powered or battery pack
[2]	GSM/GPRS-based communication	GSM-triggered SMS + buzzer	No mechanism for medicine intake monitoring logging	7-compartment model with LED indicators	Rechargeable battery
[3]	Wi-Fi + IoT backend	App + RTC module + Buzzer	No Timestamp Logging	No slot-wise indication	Battery-operated with onboard charging
[4]	No wireless module	RTC module triggers buzzer alert	No real-time monitoring, manual only	No slot-wise indication	9V battery
[5]	Not explicitly technical (pilot study)	Pre-programmed alerts via device	NA	NA	NA
[6]	No wireless module	RTC with buzzer + LCD reminder	No auto-logging, user-checked	No slot-wise indication	Rechargeable Li-ion battery
*****	Proposed Work				

5.Conclusion

This work has presented a Bluetooth-enabled Smart Pill Box designed to support medication adherence through real-time scheduling, slot-wise interaction logging, and mobile-assisted configuration. By employing a modular firmware architecture with non-blocking routines and interrupt-driven inputs, the system demonstrated reliable performance with minimal latency and efficient power consumption. Validation confirmed its robustness in alarm triggering, user interaction detection, and Bluetooth communication under typical home conditions. Beyond its technical performance, the system highlights how low-cost embedded platforms can evolve into ambient intelligence nodes that actively support independent living and aging in place. By externalizing memory through visual, auditory, and interaction-based cues, the design reduces the cognitive burden on users and ensures more consistent medication adherence. If supplemented with usability feedback, this approach can be further validated as a practical assistive technology for elderly and chronically ill populations. Nevertheless, certain limitations remain: susceptibility to mechanical switch bounce in noisy environments, reliance on visual and mechanical cues that cannot fully guarantee pill ingestion, and additional power drain during prolonged buzzer activity. Future work will focus on enhancing reliability and ecosystem integration. Planned improvements include incorporating multimodal sensing (e.g., capacitive, infrared, weight-based, or camera-assisted confirmation), implementing secure Bluetooth data transmission, and extending the system into cloud- or caregiver-linked telehealth platforms. By situating the Smart Pill Box within a larger smart home healthcare ecosystem, these advancements will strengthen its role as a resilient, context-aware, and user-centered assistive technology for next-generation ambient living environments. This work contributes toward the broader vision of resilient, user-centered, and interoperable ambient intelligence ecosystems for home healthcare.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

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